Status	Finished
Started	Thursday, 1 October 2026, 11:39 AM
Completed	Thursday, 1 October 2026, 11:43 AM
Duration	3 mins 49 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It complicates the analysis process
- c. It makes the analysis slower
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It bubbles up the largest element in each pass
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. Each part is sorted independently
c. The parts are merged without sorting
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting decreases the efficiency of selection operations
c. Sorting helps in finding duplicates quickly
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **5**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It allows for faster searching
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **6**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. Selection
c. Duplicates
d. All of the above

The correct answer is: All of the above

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Rearranging
c. Merging
d. Sorting

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Bubble Sort
- c. Quick Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are compared element by element
- b. They are split again into smaller sublists
- c. They are combined to form a single sorted list
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. order()
c. sort()
d. sorted()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is less efficient than bubble sort
c. It does not require recursion
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Heap Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Bubble Sort